WORKED SOLUTION

STEP I 2014, Question 7

Pure • Vectors in a Triangle

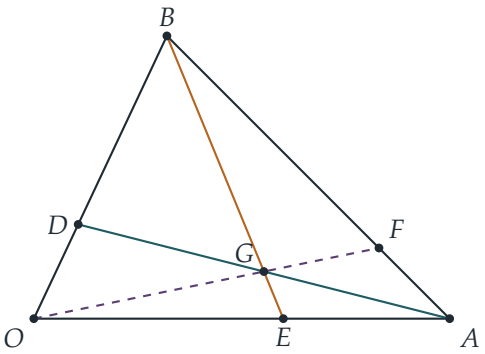
STEP I 2014, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/14-s1-q7

The key idea

Write every point as a combination of **a** and **b**. A point on a line through two known points has a one-parameter vector equation, so G can be written in two ways, one from each line. Because **a** and **b** are not parallel (the triangle is non-degenerate), we may compare coefficients, which gives two simple equations for the two parameters.



Solution

First we locate D and E. Since $BD = r\,DO$, the point D is the fraction $\frac{1}{1+r}$ of the way from O to B; since $OE = s\,EA$, the point E is the fraction $\frac{s}{1+s}$ of the way from O to A. So

$$\mathbf{d} = \frac{1}{1+r}\,\mathbf{b}, \qquad \mathbf{e} = \frac{s}{1+s}\,\mathbf{a}.$$

Part (i)

A general point of the line AD is

$$\mathbf{a} + \lambda(\mathbf{d} - \mathbf{a}) = (1-\lambda)\,\mathbf{a} + \frac{\lambda}{1+r}\,\mathbf{b},$$

and a general point of the line BE is

$$\mathbf{b} + \mu(\mathbf{e} - \mathbf{b}) = \frac{\mu s}{1+s}\,\mathbf{a} + (1-\mu)\,\mathbf{b}.$$

At G these are equal.

Ask yourself Why are we allowed to equate the coefficients of **a** and of **b** separately?

Because O, A, B form a genuine triangle, **a** and **b** are non-zero and not parallel. If $\alpha\mathbf{a} + \beta\mathbf{b} = \alpha'\mathbf{a} + \beta'\mathbf{b}$ with $\alpha \neq \alpha'$, then **a** would be a multiple of **b**, which is impossible. So

$$1 - \lambda = \frac{\mu s}{1+s}, \qquad \frac{\lambda}{1+r} = 1 - \mu.$$

The second equation gives $\lambda = (1+r)(1-\mu)$. Substituting into the first,

$$\begin{aligned} 1 - (1+r)(1-\mu) &= \frac{\mu s}{1+s} \\ \mu\left[(1+r) - \frac{s}{1+s}\right] &= r \\ \mu\,\frac{(1+r)(1+s) - s}{1+s} &= r \\ \mu\,\frac{1+r+rs}{1+s} &= r, \end{aligned}$$

so

$$\mu = \frac{r(1+s)}{1+r+rs}.$$

This is allowed because $1 + r + rs > 0$, as $r, s > 0$. Putting this value of μ into the equation of BE, the coefficient of **a** is

$$\frac{\mu s}{1+s} = \frac{rs}{1+r+rs},$$

and the coefficient of **b** is

$$1 - \mu = \frac{1+r+rs-r-rs}{1+r+rs} = \frac{1}{1+r+rs}.$$

Hence

$$\mathbf{g} = \frac{rs}{1+r+rs}\,\mathbf{a} + \frac{1}{1+r+rs}\,\mathbf{b},$$

as required.

Part (ii)

Every point of the line OG is a scalar multiple $k\mathbf{g}$ of **g**, since the line passes through the origin:

$$k\mathbf{g} = \frac{k}{1+r+rs}\,(rs\,\mathbf{a} + \mathbf{b}).$$

A point $\alpha\mathbf{a}+\beta\mathbf{b}$ lies on the line AB exactly when $\alpha+\beta=1$, because AB consists of the points $(1-\nu)\mathbf{a}+\nu\mathbf{b}$. So at F we need $k(rs+1)=1+r+rs$, and

$$\mathbf{f}=\frac{rs}{1+rs}\,\mathbf{a}+\frac{1}{1+rs}\,\mathbf{b}.$$

On the other hand, $AF=t\,FB$ means that F is the fraction $\frac{t}{1+t}$ of the way from A to B:

$$\mathbf{f}=\frac{1}{1+t}\,\mathbf{a}+\frac{t}{1+t}\,\mathbf{b}.$$

Comparing the coefficients of $\mathbf{a}$ (allowed for the same reason as before) gives $1+t=\dfrac{1+rs}{rs}$, so

$$t=\frac{1}{rs}.$$

The coefficients of $\mathbf{b}$ agree too: $\dfrac{t}{1+t}=\dfrac{1/(rs)}{(1+rs)/(rs)}=\dfrac{1}{1+rs}$.

Checks

- **Medians.** With $r=s=1$, D and E are midpoints, so G should be the centroid. The formula gives $\mathbf{g}=\frac{1}{3}\mathbf{a}+\frac{1}{3}\mathbf{b}$, which is right, and $t=1$, so F is the midpoint of AB, as it must be, since the three medians are concurrent.
- **Ceva's theorem.** The answer says $r\cdot s\cdot t=1$, that is, $\dfrac{BD}{DO}\cdot\dfrac{OE}{EA}\cdot\dfrac{AF}{FB}=1$, which is exactly Ceva's theorem for the concurrent lines AD, BE, OF.
- **A numerical case.** Take $\mathbf{a}=(5,0)$, $\mathbf{b}=(0,5)$, $r=2$, $s=3$. Then $D=(0,\frac{5}{3})$, $E=(\frac{15}{4},0)$ and the formula gives $G=(\frac{30}{9},\frac{5}{9})=(\frac{10}{3},\frac{5}{9})$. This lies on AD, whose equation is $x+3y=5$, and on BE, whose equation is $4x+3y=15$. The line OG meets $x+y=5$ at $F=(\frac{30}{7},\frac{5}{7})$, and $\overrightarrow{AF}=(-\frac{5}{7},\frac{5}{7})$, $\overrightarrow{FB}=(-\frac{30}{7},\frac{30}{7})$, so $AF:FB=1:6$, giving $t=\frac{1}{6}=\frac{1}{rs}$. The formulas were also confirmed by computer for two thousand random triangles and ratios, using exact arithmetic.

Common slips

- Getting the ratios the wrong way round: $BD=r\,DO$ puts D at $\frac{1}{1+r}$ of the way from O to B, not $\frac{r}{1+r}$.
- Comparing coefficients without saying why it is valid: it needs $\mathbf{a}$ and $\mathbf{b}$ to be non-parallel.
- In part (ii), forgetting to rescale $\mathbf{g}$: G itself is not on AB, since its coefficients add up to less than 1.

Beyond STEP

Coefficients that add up to 1, like those of **f**, are *barycentric coordinates*. They describe points by weights at the vertices, and they make results such as Ceva's and Menelaus' theorems almost automatic. In the language of linear algebra, comparing coefficients works because **a** and **b** are linearly independent and so form a basis of the plane. Strang's *Introduction to Linear Algebra* develops bases and coordinates carefully.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.